

2.5.1 Title, researchers and institution

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JouleFlex: A Model-Based Decision and Control Framework for Maximizing Useful AI Work under Dynamic Power Limits

JouleFlex: מודלים ובקרה חכמה למקסום תפוקת AI תחת מגבלות הספק משתנות

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2.5.2 Abstract / תקציר

English (299/300 words)

In July 2026 Israel's Electricity Authority froze new data-center connection requests after the queue reached 27,000 megawatts. Under electricity limits, data centers switch GPUs (AI processors) off or cap their power, deciding in watts (the momentary draw), not joules (the energy consumed). An over-capped GPU computes slower and pays its fixed power floor longer, so energy per unit of useful work rises even as the meter shows fewer watts: less AI per kilowatt-hour. Aviation solved the same problem: fuel flow rises far faster than speed, yet slow flight burns fuel longer, so airlines cruise at the speed minimizing fuel per kilometer.

The PI's study [1] measured the time-power tradeoff for AI workloads (5,500 measurements, 20 models, four GPUs). Energy per task is U-shaped in the power cap. On the large GPUs its minimum, the Joule Point (JP), sits at 43 to 46 percent of rated power; holding a card there cut energy per task 29 to 31 percent while running 1.2 times slower. Even under a strict deadline, capping saved 28 percent of energy while keeping 95 percent of workload value. A fleet simulation on the measured curves served equal work with 18 to 45 percent less energy: about 50 percent more AI work per megawatt.

These insights change how AI data centers are planned, managed, and priced, raising decisions no current tool supports: how to design a flexible-connection-ready fleet whose cards run near their JP; how to measure and predict the JP for new cards and jobs; and how to price service to motivate customers toward efficiency, not per-hour billing rewarding the energy-worst point. Each is a modeling problem. This project delivers the decision-support toolkit answering all three, released with an open measurement dataset. Run near its JP, a fleet frees curtailable connection capacity for Israel's flexible-connection scheme.

עברית (300/300 מילים)

ביולי 2026 הקפיאה רשות החשמל בקשות חיבור חדשות של חוות שרתים (מרכזי נתונים) לרשת החשמל, לאחר שהתור הגיע ל-27,000 מגה-ואט. כאשר אספקת החשמל מוגבלת, חוות שרתים מכבות מעבדי GPU או קובעות להם תקרת הספק, ומחליטות בוואטים (ההספק הרגעי) ולא בג'ולים (האנרגיה הנצרכת). מעבד

GPU שתקרת ההספק שלו נמוכה מדי מחשב לאט יותר וצורך את הספק הסרק שלו לאורך זמן, ולכן האנרגיה ליחידת חישוב מועיל עולה אף שקריאת ההספק יורדת: פחות בינה מלאכותית לכל קילואט-שעה. התעופה פתרה אתגר דומה: צריכת הדלק עולה מהר מן המהירות, אך טיסה איטית שורפת דלק זמן רב יותר, ולכן משייטים במהירות הממוצעת דלק לקילומטר.

מחקר שערך החוקר הראשי [1] מדד את הפשרה בין זמן להספק בעומסי עבודה של בינה מלאכותית (5,500 מדידות, 20 מודלים, ארבעה מעבדי GPU). האנרגיה למשימה מתארת עקומת U כתלות בתקרת ההספק. בדגמי ה-GPU הגדולים נקודת המינימום, נקודת הגיול (Joule Point; JP), מצויה ב-43 עד 46 אחוזים מההספק הנקוב; הפעלת מעבד GPU בנקודה זו הפחיתה את האנרגיה למשימה ב-29 עד 31 אחוזים בעוד הריצה התארכה פי 1.2. אף תחת מגבלת זמן קשיחה, הגבלת ההספק חסכה 28 אחוזים מהאנרגיה תוך שמירת 95 אחוזים מערך העבודה. סימולציה של צי מעבדים על העקומות שנמדדו סיפקה עבודה שווה בצריכת אנרגיה נמוכה ב-18 עד 45 אחוזים: כ-50 אחוזים יותר עבודת בינה מלאכותית לכל מגה-ואט.

תובנות אלה משנות את תכנון, ניהול ותמחור חוות השרתים לבינה מלאכותית, ומעלות החלטות שאין להן כלי תומך: כיצד לתכנן צי המותאם להסדר חיבור גמיש שמעבדי ה-GPU שלו פועלים סמוך ל-JP; כיצד למדוד ולחזות את ה-JP עבור משימות ומעבדי GPU חדשים; וכיצד לתמחר את השירות כך שיתמרץ לקוחות להתייעלות באנרגיה, במקום חיוב שעת המתגמל את נקודת ההפעלה הבזבזנית ביותר. כל אחת מהן היא בעיית מידול. הפרויקט יספק ערכת כלים תומכת החלטות העונה על שלושתן, שתשחרר עם מאגר מדידות פתוח. צי הפועל סמוך ל-JP משחרר קיבולת חיבור הניתנת להפחתה עבור הסדר החיבור הגמיש של רשות החשמל.

2.5.3 Glossary

Term	Meaning
Power	How fast electricity is drawn at a given moment; watts (W) or megawatts (MW). The grid connection is sized by peak power.
Energy	The total electricity consumed: power $\times$ time; joules or kilowatt-hours. The bill and the carbon footprint are set by energy.
Power cap	A software limit on a GPU's power draw. Set with one command; takes effect in ~ 0.2 seconds. The control knob of this project.
TDP	A GPU's full rated power. Running at TDP is today's default.
Joule Point (JP)	The power cap at which energy per task is lowest for a given card; measured at 43–46% of TDP on large GPUs.
Operating point	The pair (GPU, power cap) a job runs at.
Power envelope	A limit on a facility's total power draw that can change over time, set by a budget, a price or carbon signal, or a grid instruction.
Power shaping	Keeping a fleet inside a power envelope by capping jobs, like traffic shaping in networks.
ELF	Our released dataset: dense power-cap sweeps of 20 AI models on 4 GPUs, $\sim 5,500$ measured rows.
SLO	Service level objective: the deadline or latency a job must meet.

Firm / flexible capacity	Grid connection capacity guaranteed at all times, versus capacity given on condition that the facility reduces power when instructed.
Card multiplier	The extra cards needed to keep total throughput when capping. It equals the latency ratio ($\sim 1.2\times$).

2.5.4 Objectives of the program

2.5.4.1 The objective

The Joule Point. A GPU's energy per unit of work is not fixed. As its power cap is lowered from the factory maximum, throughput falls more slowly than power, so energy spent per task drops to a minimum, the **Joule Point (JP)**, before rising again: the cap at which a card does the most AI work per joule. The PI's study [1] measured the JP across 20 models and 4 GPUs (three swept finely); it sits well below the thermal design power (TDP), is near-constant per card, and delivers 29–31 percent less energy per task at about 1.2 times the runtime. Production data centers still run cards at maximum power, and Israel's grid is being sized for that maximum. Capturing the JP, and the load flexibility it exposes, is what this project turns into deployable tools.

The objective is to construct **JouleFlex**: a toolkit of models, software tools and a simulator for the energy management of GPU-rich AI data centers, built on the JP concept, in three layers. **Models:** the power-response law validated on current hardware and whole servers; a job model calibrating the energy-optimal cap per workload; a transfer model for unmeasured GPUs; and a joint job-and-cap assignment method. **Tools:** a characterization kit qualifying a new GPU within the validated hardware domain in hours, and a fleet control engine holding a facility inside its power envelope while meeting deadlines. **Simulation:** a digital twin replaying measured curves, evaluating any policy or connection scenario before deployment. All layers are released openly with the calibrating corpus (ELF-2, the extended measurement dataset), serving two users: the operator who runs it, and the Ministry that plans, regulates and sizes connections with its numbers. The twin answers the fleet-design question directly: how many cards, of which type, run near the JP behind a given connection.

2.5.4.2 The research question

Can an AI data center run as a controllable electrical load under Israeli grid constraints: each GPU near its energy-optimal cap, the facility inside a changing power limit, less energy and less connected power per unit of AI work, and every admitted job still meeting its deadline?

Six sub-questions structure the work plan:

1. **Does the law generalize?** Do the power law and JP (measured on 20 models, 4 GPUs; three swept finely) hold on H100/L40S-class hardware (B200-class as access allows), production serving software, and shared GPUs?
2. **When is one cap per card enough,** and where must a job model replace the near-constant per-card JP?
3. **Where is the true optimum at the wall?** What is the JP at server-input power, which the grid meters, not just the board?

4. **Can we skip the measurement?** Can an unmeasured GPU's JP be predicted from published specs plus at most one short test?
5. **Can allocation and power be optimized together, in motion?** Can job placement and per-card caps be chosen jointly under a time-varying power limit with every deadline met? (The measured result covers fixed caps and placement; the joint problem is open.)
6. **How much flexibility can a facility sell, and who has the incentive?** How much power can be shed, how fast, for how long, at what cost, and which pricing rules make flexibility profitable?

2.5.5 The proposed research

2.5.5.1 Scientific and technological background

The Israeli challenge is specific and pressing. Data centers, with AI the main growth driver, approach 945 TWh of electricity demand by 2030 [3], and Israel's grid is an isolated island system with no cross-border interconnections, so new demand must be met from domestic capacity; in July 2026 the Electricity Authority halted new data-center connection requests for 140 days, after the queue reached about 27,000 MW, roughly three times average national consumption [2]. The binding limit is grid capacity in megawatts: the national need is more AI work from every connected megawatt.

Today's operating decisions think in watts, not joules. Under stress a data center switches GPUs off, discarding service, or caps power energy-blind: a capped GPU runs slower and pays its fixed power floor longer, so capping too deep lowers the dashboard watts while raising the joules on the bill, and unconstrained the default is full rated power, where the last increment of speed carries a steep surcharge.

The discovery. The PI's study [1] mapped, over 20 AI models and four GPU types (~5,500 instrumented measurements, released as the ELF dataset), how energy per task depends on the power cap: board power follows a simple empirical law in the computing rate θ ,

$$P(\theta) = P_0 + a \cdot \theta^\beta, \quad \text{median } R^2 = 0.99,$$

a fixed floor P_0 plus a superlinear term. Energy per task, $E = P/\theta$, is therefore U-shaped in the cap; its minimum, the **JP**, sits at 43 to 46 percent of rated power on the large GPUs, cutting energy per task by 29 to 31 percent at an exactly priced cost: tasks run about 1.2 times slower, and holding throughput takes that factor more cards. The cap is one software command, settling in about 0.2 seconds, fast enough to follow grid signals; behind a fixed feed the measured A100 curve implies about 50 percent more AI tasks per megawatt [1].

2.5.5.2 Review of existing knowledge

Four research lines meet here. *Energy-proportional computing* [4] identified the fixed, work-independent part of server power as the central inefficiency; our floor P_0 measures it for modern GPUs, and the JP is where paying it stops being worthwhile. *Speed-scaling theory* proved that energy is minimized at a sub-maximal speed [5,6,7]; the ELF dataset gives the measured form for GPU inference, where the law holds per operating point and loosens to R^2 0.88–0.92 when pooled across workloads. *GPU energy systems*, notably Zeus [8] and the ML.ENERGY benchmarks [9,10], search for a good setting per job at runtime; the per-card-constant JP replaces that search with a one-time characterization. *Production context*: cluster studies [11] and public traces [12] supply realistic job arrivals; energy-aware GPU schedulers [13,14] leave each device's power draw at its default; methodology reviews [15] motivate our fixed protocol. Recent work power-caps GPU inference servers for energy savings [20] but reports a single recommended cap, not the optimum's measured location across many models and GPUs, and one analysis finds capping inert for memory-bound decode [23], a regime ELF's loaded-serving sweeps bound. AI data centers have been demonstrated as grid assets [21] by following a facility power target and cutting low-priority work first; JouleFlex instead runs each job at its measured energy optimum and states the firm and flexible capacity in MW, ramp time and recovery time.

2.5.5.3 Knowledge gaps

- **The law's reach is unmapped.** It is established for single jobs at fixed settings on the three finely swept GPUs (A100, A10G, L4); its persistence on H100 and, as access allows, B200-class hardware, production serving software and shared cards is unknown, and placing the JP on a new GPU still requires a full sweep [1].
- **The job-dependence structure is unmapped.** The per-card constant holds in the measured loaded regime; outside it the optimum moves, and no model predicts a job's operating point from its features.
- **All published numbers are board numbers.** The grid meters the wall; host overhead (CPU, memory, cooling, power-supply losses) shifts the true optimum upward by an unmeasured amount, so no megawatt claim is yet grounded.
- **Allocation and power control are not optimized together in the cited work.** The measured 18 to 45 percent fleet saving used a fixed budget, a fixed placement rule and caps chosen afterward; the joint problem, coupling discrete scheduling to continuous control through the shared power budget, is unsolved.
- **Flexibility is unquantified.** How much power an AI facility can shed, how fast, for how long, at what cost has never been measured or standardized; Israel is rationing connections without those numbers.
- **The incentive problem is identified but unsolved.** Per-hour billing makes the renter's cheapest choice the energy-worst point, and a per-joule surcharge is too weak [1]; which structural instruments work is open.

2.5.5.4 Description of the proposed research

JouleFlex builds one measured foundation and four tools on it: first work out the energy-optimal way to run each job on each card, then control power to achieve it. **WP1–WP2 measure and predict** the efficient operating point; **WP3 controls** allocation and power in real time, tested in a digital twin; **WP4–WP5 connect to the grid**, defining the flexibility a controlled facility can offer and the pricing and connection rules that reward efficient operation. Everything is released openly. The work packages below give the detail.

2.5.5.5 Methods (work packages)

WP	Objective	Approach and output
WP1	Extend the energy-versus-power measurements to current GPUs, production software, and whole-server power.	Re-run the study's measurement sweep on current GPUs (H100/L40S class) and production serving engines (vLLM [16], TensorRT-LLM), reading power at three tiers: GPU board (NVML), host CPU (RAPL), and server-input, the wall tier (BMC via IPMI/Redfish). The JP is the energy minimum, where the tangent from the origin meets $P(\theta)$ ($dE/d\theta = 0$); a meter gain error leaves its location unchanged while an additive offset shifts it, so JP location and per-card savings are gain-robust, while absolute board-to-wall and megawatt figures carry the offset caveat. <i>Output</i> : the ELF-2 open dataset (board and server-input).
WP2	Predict any new card/s or job's efficient operating point without a full measurement sweep.	A job model predicts a job's best power cap from its features; a transfer model predicts a new card's curve from published specifications plus one short test. Both report error bars and a stated range of validity, and a short proof bounds the small penalty of one cap per card. <i>Output</i> : the characterization kit (models and one-test protocol).
WP3	Decide, in real time, which job runs on which card and at what power cap, to do the most useful work within a changing power limit.	A controller that makes both choices together, since moving a job changes the power left for others, tested in a digital twin that replays measured curves against exact baselines. When power is tight it defers low-priority work and reports the cost. The joint problem is NP-hard; the per-card capping step is solved exactly. <i>Output</i> : the open control engine and digital twin.
WP4	Define what flexibility a controlled facility can offer the grid, and measure it.	Specify the contractable quantities, a committed power envelope, a flexibility offer, and an auditable delivery record, with ramp and recovery metrics; demonstrate them on a rented metered server and at fleet scale in the twin, under an Israeli firm-plus-flexible scenario. <i>Output</i> : the flexibility products and the Israel demonstration.
WP5	Find pricing and connection rules that reward efficient operation, and release everything openly.	An incentive model showing why per-hour billing rewards the energy-worst point and what changes fix it; a valuation of the flexible-connection option from public Noga and Electricity Authority data; each rule tested in the WP3 twin. <i>Output</i> : the policy report and the open release of the dataset, control engine, twin, and interfaces.

Success metrics are pre-registered per work package; the stage table in 2.5.7.1 states each threshold.

2.5.5.6 Novelty and originality

The measured law and the JP are established preliminary results [1]; the project's novelty is the frontier built on them:

- **AI load as a grid resource, grounded in the measured optimum.** Unlike grid-following curtailment to a power target [21], JouleFlex runs each card at its measured energy-optimal cap and makes the resulting flexibility contractable and audited, quantified in MW, ramp and recovery.
- **From per-job search to one-time measurement, with an exact price.** In the loaded regime, one per-card-model measurement replaces per-job runtime search at under one percent penalty [1]; extra cards to hold throughput equal the latency ratio, so every recommendation carries its exact capital cost.
- **Energy first, power as the means.** Inference power-capping saves energy at a chosen cap [20] and can be inert for memory-bound decode [23]; JouleFlex instead sets power to the computed energy optimum per (card, job) in the loaded regime where the cap binds.
- **An open dataset with no equivalent.** WattGPU [22] predicts inference power at a single operating point and ML.ENERGY [9,10] reports one point; open power-cap sweeps cover training or HPC. The intersection, a dense power-cap sweep across many inference models and GPUs at the granularity to locate the energy-optimal point, does not exist. ELF fills it.
- **The joint allocation-and-capping problem, to the measured optimum.** Placing jobs and setting every cap at the per-(card,job) energy optimum, coupled through the shared power budget, defines a scheduling-control problem grid-following controllers [21] do not address; the JouleFlex engine is a first solution driven by the measured curves.

2.5.5.7 Fit to the Ministry's R&D goals

The project is submitted under topic 16.2 (AI and energy for data centers) of the Ministry's 2026 R&D policy [17], call 34/2026 [18]; its core deliverable answers 16.2 and its methods land on three 2026 priorities (8, 16.1, 6.3). Every artifact serves a decision the Ministry faces now: the July 2026 Electricity Authority freeze of new data-center connections against a 27,000 MW queue [2], with no measured basis for how much of it can flex. Clause by clause (Chapter 1 numbering; 2026 priorities marked):

Clause	How the project addresses it
16.2 · טכנולוגיות תכנון וניהול הספקת אנרגיה לחוות שרתים	The core deliverable. The control engine manages a farm's energy supply in real time (each card at its measured optimum, the fleet held to a committed envelope); the twin sizes how many cards run behind a given connection and how much power can be shed.
8 · התייעלות באנרגיה ובזיכור מערכות אנרגיה (2026 priority)	Energy efficiency of a new industrial load class (8.1, 8.3): 29–31 percent less energy per task at the card [1] and 18–45 percent per served job at the fleet (simulation), in software on installed hardware.

16.1 · שימוש בבינה מלאכותית ככלי עזר לפיתוח ומחקר בכל הנושאים לעיל (סעיפים 1-15) (priority 2026)	Machine learning as the instrument for topics 8 and 6: the WP2 job model predicts a job's energy-optimal cap from its features and the transfer model predicts a new card's curve from its specifications plus one short test; the WP3 controller solves joint allocation-and-capping over the learned curves.
6.2 / 6.3 · ניהול אנרגיה מקומי וניהול ביקושים · בקרה וניהול רשת בחדירת מתחדשות גבוהה (6.3 : priority 2026)	Demand management: a facility holding a committed envelope and shedding on instruction is a demand-response resource; cap actuation settles in about 0.2 s at the card and the demonstration targets a 60-second facility ramp, a response time within a demand-response window, letting a system operator on an isolated grid count AI load as reserve.
15.1 / 15.3 · כלי מדיניות ליעול השימוש באנרגיה · מתודולוגיה וכלי עזר בקבלת החלטות	Policy tools for efficient energy use (15.1): WP5 tests billing and connection rules in the twin and shows which reward the energy-optimal point. Decision support (15.3): the regulator-facing report quantifies firm-plus-flexible connections against the queue.

2.5.5.8 Preliminary results

The preliminary results are the applicant's study *The Joule Point* [1] and its ELF dataset: about 5,500 instrumented measurements over 20 models and four GPUs establish the response law (median $R^2 = 0.99$), locate the JP at 43 to 46 percent of TDP with a 29 to 31 percent energy saving per task at 1.2 times longer runs, show one static cap per card serving all twenty workloads at under one percent mean penalty, and demonstrate an 18 to 45 percent fleet-level saving at equal-or-better deadline compliance in a measured-curve simulation. Cap actuation settles in about 191 milliseconds. Under the per-instance-hour billing model analyzed in [1], the renter's cost optimum is the energy-worst operating point. ELF spans four GPU types; the fine sweeps behind the law and the JP cover three (A100, A10G, L4), with the T4 swept coarsely for coverage, and $R^2 = 0.99$ is the median per fixed-setting sweep while pooled per-card fits across workloads are 0.88 to 0.92 (Fig. A1). Under the strict-deadline experiment of [1], capping saved 28 percent of energy while keeping 95 percent of workload value.

2.5.6 Application: expected benefit and impact

2.5.6.1 What the project delivers, concretely

The project delivers five artifacts: the ELF-2 open dataset, the characterization kit, the control engine, the digital twin, and the grid interfaces and policy report. Appendix Table A1 states what each is and who uses it. Each artifact is released openly on delivery, every release crediting the Ministry's R&D program under call 34/2026 [18].

2.5.6.2 Benefit and impact

For the Israeli grid and regulator (the primary benefit). Under the July 2026 freeze [2], Israel faces a binary choice: deny AI-load growth, or overbuild firm generation to serve it. JouleFlex supplies a third path. The binding constraint is connection capacity in megawatts, and the JP converts efficiency into capacity: on the measured A100 curve, a fixed-feed calculation implies

about 50 percent more AI work per megawatt of GPU power [1]. Every 1,000 MW of the 27,000 MW queue connected under JP operation could carry the AI work of up to 1,500 MW uncapped, a board-level figure WP1 and WP4 verify at server input and facility level before any coefficient is recommended. WP5 puts a shekel-per-megawatt value on the flexible-connection option from public Noga and Electricity Authority data, entering measured numbers into the Authority's open consultation [19] while the rules are written. With flexibility measured, committed and audited (WP4), the regulator gains a third option and the frozen queue stops being all-or-nothing.

Training and human capital (an explicit impact objective). Israel has little research capacity at the AI-energy interface, which operators and the regulator now both need. The project builds it on three levels: a postdoctoral researcher trained across a stack no single Israeli group combines (GPU energy measurement with NVML/RAPL/BMC, ML-systems control, and energy-market analysis); the open ELF-2 dataset, twin and control engine as teaching material at HIT; and the Hebrew M8 and M21 briefings and public release that carry the methods to Ministry, Authority and Noga staff and to operators. A trained specialist, an open national dataset and a better-equipped regulator outlast the grant.

For operators. On installed hardware, with no new equipment, one software command delivers 29 to 31 percent lower energy per task (measured) and 18 to 45 percent lower per served job (fleet simulation).

2.5.7 Work plan

2.5.7.1 Stages, deliverables, duration and success metrics

Months	Stage	Deliverable	Success metric
M1–M6	WP1: harness; new-generation and production-engine (vLLM) sweeps; board and node instrumentation	D1: ELF-2 v1	Law fit $R^2 \geq 0.95$ per sweep (measured median 0.99 [1]); any platform where it fails is documented as a bounding result; JP located (or absence documented) per platform at board and node power
M5–M8	WP1: shared GPUs, LLM phases, fine-tuning; sweeps where the constant should break	MS1: regime map	Server-input (BMC) wall tier validated; validity domain documented, including the pre-registered JP shift (percentage points of TDP) at which the per-card constant breaks
M7–M12	WP2: job model; transfer model; one-test characterization protocol	MS2: models result	Job-model placement within 5 percentage points of TDP where the constant breaks; transfer-model placement within 5 percentage points of TDP on held-out cards; protocol cost ≤ 8 hours and $\leq \text{₪}1,500$ per card model

M9– M16	WP3: twin; joint controller under fixed, price, carbon and curtailment envelopes	MS3: controller result	Energy per served job and SLO compliance versus uncapped, static-cap and cap-only baselines; a pre-registered floor of at least 10 percent below cap-only at equal compliance (incremental gain over an already-capped fleet, not over full power); zero exceedance beyond the WP1 noise band in committed windows, transient violations (under 1 percent of control intervals) reported
M14– M20	WP4: flexibility products and metrics; demonstration on rented BMC-metered servers; Israel scenario	MS4: flexibility demonstration	Sheddable power as a fraction of the committed envelope (pre-registered floor: at least 15 percent within the ramp deadline at bounded SLO loss), measured on a BMC-metered server, projected to fleet scale in the twin; ramp within 60 seconds and recovery within 5 minutes, a response time compatible with the curtailment right proposed in the Authority's hearing [19]; delivery-record compliance and service cost
M17– M21	WP5: incentive model; connection valuation on published Israeli data; twin evaluation	MS5: policy report draft	Alignment condition per billing structure, confirmed in the twin; option value per MW and break-even curtailment frequency, with sensitivity to every price input
M21– M24	WP5: release, documentation, validation, publication	D2: open release; D3: final report	Public repositories with reproducing pipelines; metered validation; final scientific and policy reports

Two interim briefings are named deliverables: the M8 technical brief and the M21 policy draft, in Hebrew, circulable to the Authority and Noga [19]. Each milestone is tested against pre-stated criteria, and a bounding result is itself a deliverable. Minimum-success path: H100/L40S validation, one serving engine, BMC wall measurement, the job and transfer models, the joint controller, and one Israeli flexibility scenario; B200-class cards, a second engine, fine-tuning, shared-GPU regimes are extensions as access allows.

2.5.7.2 Budget by stage

Stage	Share	Amount (₪)	Principal costs
M1–M8 (WP1; WP2 begins)	30%	148,314	Postdoctoral researcher; GPU rentals for cap sweeps
M9–M16 (WP2 ends; WP3; WP4 begins)	33%	163,145	Postdoctoral researcher; card-model rentals for transfer validation; twin and controller work
M17–M21 (WP4 ends; WP5)	25%	123,595	Postdoctoral researcher; BMC-metered rentals; flexibility campaign
M22–M24 (WP5: release and validation)	12%	59,326	Postdoctoral researcher; metered validation runs; documentation; dissemination and publication
Total		494,380	Full itemization in the מפרט תקציבי

2.5.7.3 Risks and mitigation

Risk	Impact	Mitigation
Production serving software (continuous batching, shifting phases) blurs the per-card constant	Static cap loses near-optimality for LLMs	Measured first (WP1, M1); the job model predicts the optimum where the constant breaks
Power-cap control unavailable on some rented platforms	Measurement campaign narrowed	AWS EC2 gives passthrough and root; too-high cap floors are exposed via the graphics clock (L4 [1]); BMC rentals cover the wall tier
Postdoc recruitment slips, or the single hire is lost mid-project	Schedule and delivery at risk	PI runs the WP1 bench directly (the ELF harness is his); the minimum-success path is one-FTE executable; recruitment via HIT and national channels with a fallback
Server-input power readout (BMC via IPMI/Redfish) blocked on a rented server	Wall-tier measurement narrowed	Named BMC-capable bare-metal vendors with dated quotes (GPU Mart and equivalents); fallback to an in-line metered PDU at the wall

2.5.8 The applicant

Project manager and principal investigator: Alexander Apartsin, Senior Lecturer, Computer Science, HIT; single-PI, responsible for all five work packages, committing ~one day per week (~20 percent) over 24 months, no grant salary (\$2.6).

Fit of expertise to the proposed work. The project builds on the PI's own research: study [1] produced the ELF dataset, the power law and the Joule Point, and its harness, models and simulator are working code that WP1–WP3 extend. The PI's background fits the wider plan on both sides: an academic thread in energy-efficient AI, spanning energy-efficient remote-sensing measurement [24] and the recent CalexNet study of efficient inference in deep models [25]; and an industry thread in cost-aware AI systems and capacity-and-cost engineering, including system-sizing at Amdocs that contributed about \$10M in savings. That mix of measurement-driven machine-learning research and cost-and-systems engineering is the expertise the work plan calls for.

Research personnel. One postdoctoral researcher at 90% position, both years, carries the measurement, modeling and control work under the PI's supervision, reported within a month of signature (2.5\$ ט"ו). Core profile at hiring: GPU systems and energy instrumentation (NVML/RAPL/BMC), ML systems, and optimization or control. WP4 and WP5 are public-data modeling on Noga and Electricity Authority figures, done under the PI's direction, needing no dedicated power-systems hire. If hiring slips, the PI runs the WP1 bench directly (the ELF harness is his), preserving the minimum-success path.

Advisor. Prof. Moshe Tshuva (Afeka College of Engineering), a senior energy-systems engineer with two decades in energy-systems management and efficiency, advises the WP5

techno-economic and connection-valuation analysis and the energy basis of the impact case.

Resources. The project owns no hardware: measurement and the WP4 demo run on rented AWS and BMC-metered servers. HIT provides lab space, computing and administration.

2.5.9 Funding

No other, complementary, or parallel student funding is taken or sought (2.10\$,1.5.2\$ (נספח טז').

2.5.10 Bibliography

1. Apartsin, A. and Aperstein, Y. (2026). *The Joule Point: an Energy-Optimal Operating Point for AI Inference*. arXiv preprint, September 2026. arXiv: [2609.27926](https://arxiv.org/abs/2609.27926) (CC BY 4.0), with the released ELF dataset.
2. JNS (22 July 2026). *Israel halts new data-center grid requests for 140 days amid surge in demand*. Press report on the Electricity Authority's temporary order: pending applications of approximately 27,000 MW, about three times Israel's average electricity consumption. <https://www.jns.org/news/israel-news/israel-halts-new-data-center-grid-requests-for-140-days-amid-surge-in-demand> (the Authority's order is published on gov.il).
3. International Energy Agency (2025). *Energy and AI: Energy Demand from AI*. <https://www.iea.org/reports/energy-and-ai/energy-demand-from-ai>
4. Barroso, L. A. and Hözl, U. (2007). *The Case for Energy-Proportional Computing*. IEEE Computer 40(12), pp. 33–37.
5. Yao, F., Demers, A., and Shenker, S. (1995). *A Scheduling Model for Reduced CPU Energy*. Proceedings of FOCS 1995, pp. 374–382.
6. Miyoshi, A., Lefurgy, C., Van Hensbergen, E., Rajamony, R., and Rajkumar, R. (2002). *Critical Power Slope: Understanding the Runtime Effects of Frequency Scaling*. Proceedings of ICS 2002, pp. 35–44.
7. Gandhi, A., Harchol-Balter, M., Das, R., and Lefurgy, C. (2009). *Optimal Power Allocation in Server Farms*. Proceedings of ACM SIGMETRICS 2009, pp. 157–168.
8. You, J., Chung, J.-W., and Chowdhury, M. (2022). *Zeus: Understanding and Optimizing GPU Energy Consumption of DNN Training*. arXiv: 2208.06102. <https://arxiv.org/abs/2208.06102>
9. Chung, J.-W., Ma, J. J., Wu, R., et al. (2025). *The ML.ENERGY Benchmark: Toward Automated Inference Energy Measurement and Optimization*. arXiv: 2505.06371. <https://arxiv.org/abs/2505.06371>
10. Chung, J.-W., Wu, R., Ma, J. J., et al. (2026). *Where Do the Joules Go? Diagnosing Inference Energy Consumption*. arXiv: 2601.22076. <https://arxiv.org/abs/2601.22076>
11. Hu, Q., Sun, P., Yan, S., Wen, Y., et al. (2021). *Characterization and Prediction of Deep Learning Workloads in Large-Scale GPU Datacenters*. arXiv: 2109.01313. <https://arxiv.org/abs/2109.01313>
12. Alibaba Group (2020). *cluster-trace-gpu-v2020: production GPU cluster traces from Alibaba PAI*. <https://github.com/alibaba/clusterdata>
13. Lipe, E., Karia, N., Espenshade, C., Stein, C., Tantawi, A., and Tardieu, O. (2026). *Energy Efficient Scheduling of AI/ML Workloads on Multi-Instance GPUs with Dynamic Repartitioning*. arXiv: 2606.25082. <https://arxiv.org/abs/2606.25082>
14. Lu, T. and Reda, S. (2026). *Agentic CPU-GPU Scheduling for Heterogeneous AI Workloads*. arXiv: 2607.22242. <https://arxiv.org/abs/2607.22242>
15. Rodriguez, C., Degioanni, L., Kamení, L., Vidal, R., et al. (2024). *Evaluating the Energy Consumption of Machine Learning: Systematic Literature Review and Experiments*. arXiv: 2408.15128. <https://arxiv.org/abs/2408.15128>
16. Kwon, W., Li, Z., Zhuang, S., et al. (2023). *Efficient Memory Management for Large Language Model Serving with PagedAttention*. Proceedings of SOSPP 2023. arXiv: 2309.06180. <https://arxiv.org/abs/2309.06180>
17. Ministry of Energy and Infrastructure (2026). *R&D Support Policy for 2026* (in Hebrew). Chapter 1 clauses 6, 8, 15, 16; Chapter 2 priority list.

18. Ministry of Energy and Infrastructure (2026). *Call for Proposals 34/2026 for Research Funding in Energy and Earth and Marine Sciences* (in Hebrew).
19. Calcalist (June 2026). Report on the Electricity Authority's public hearing: a dedicated connection track for transmission-grid consumers of 50 MVA and above, including a system-operator right to curtail consumption at peak and payments for holding connection capacity. Under public consultation; cited as regulatory direction, not settled regulation. <https://www.calcalist.co.il/calcalistech/article/rylkxemgx> (in Hebrew; the hearing documents are published on gov.il).
20. Ma, Y., Srinivasan, S., and Wang, X. (2025). *Power Capping of GPU Servers for Machine Learning Inference Optimization*. Proceedings of ICPP 2025 (54th International Conference on Parallel Processing). DOI: [10.1145/3754598.3754670](https://doi.org/10.1145/3754598.3754670).
21. Colangelo, P., Coskun, A. K., Megrue, J., et al. (2025). *AI data centres as grid-interactive assets*. Nature Energy. DOI: [10.1038/s41560-025-01927-1](https://doi.org/10.1038/s41560-025-01927-1); field demonstration in Phoenix, AZ, arXiv: 2507.00909.
22. Fadel Argerich, M., Fürst, J., and Patiño-Martínez, M. (2026). *WattGPU: Predicting Inference Power and Latency on Unseen GPUs and LLMs*. arXiv: 2607.02391. <https://arxiv.org/abs/2607.02391>
23. Ma, B., Afzal, A., Eitzinger, J., and Wellein, G. (2026). *The Illusion of Power Capping in LLM Decode: A Phase-Aware Energy Characterisation*. arXiv: 2605.11999. <https://arxiv.org/abs/2605.11999>
24. Apartsin, A., Cooper, L. N., and Intrator, N. (2014). *Energy-Efficient Time-of-Flight Estimation in the Presence of Outliers*. IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing 7(4). DOI: [10.1109/JSTARS.2013.2295324](https://doi.org/10.1109/JSTARS.2013.2295324).
25. Aperstein, Y. and Apartsin, A. (2026). *CalexNet: Soft Cascade-Aligned Training and Calibration for Lightweight Early-Exit Branches*. Electronics 15. DOI: [10.3390/electronics15102149](https://doi.org/10.3390/electronics15102149).

Appendix A: Figures (within the 5-page allowance of 2.2§ 1'א'נספח)

JouleFlex · 34/2026 קורא · Appendix to the research proposal, permitted under 2.2§ 1'א'נספח (graphs, tables and diagrams, up to 5 A4 pages, outside the 10-page limit). All figures are measured results from the applicant's study [1].

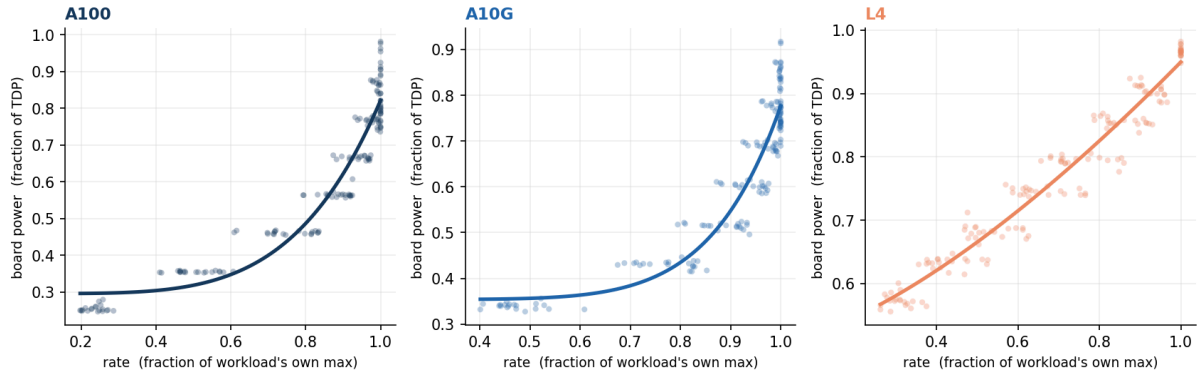


Figure A1. The response law is a card property. One panel per GPU (A100, A10G, L4); each dot is a measured operating point for one of the 20 workloads in the loaded regime, rate normalized to the workload's own maximum and power to the card's TDP. All workloads follow a single fitted law per card, $P(\theta) = P_0 + a \cdot \theta^\beta$ (pooled per-card R^2 of 0.92, 0.88 and higher per-sweep), showing the exponent travels with the card rather than the model. This is the empirical basis for the one-time per-card characterization that WP2 turns into a transfer model.

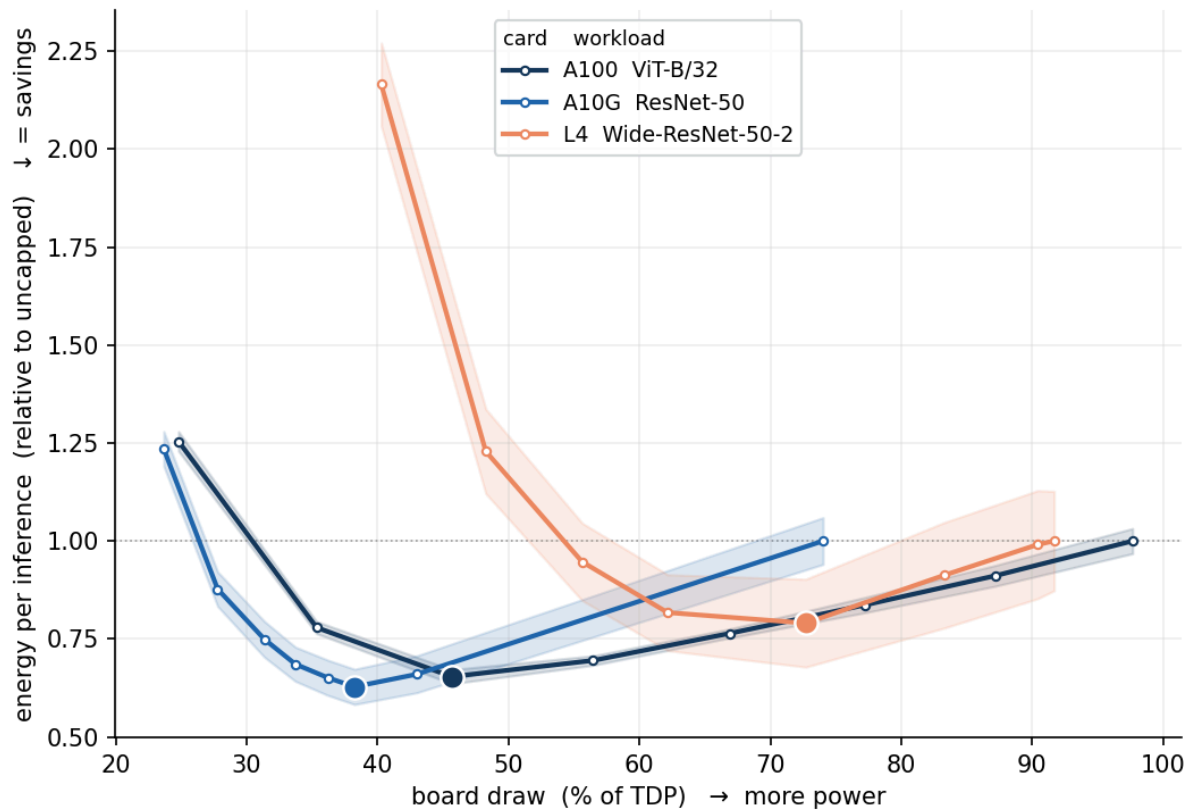


Figure A2. Every card has an energy U-shape. Energy per inference, normalized to each card's own uncapped value, against board draw as a fraction of TDP; the dot marks the JP, the shaded band one standard deviation over repetitions. On the large GPUs the minimum sits at 43–46 percent of TDP and capping to it cuts energy per inference by 29–31 percent; on the L4 the interior minimum lies below the power cap's reachable floor and is exposed by the graphics clock instead, mapping which actuator applies to which card class.

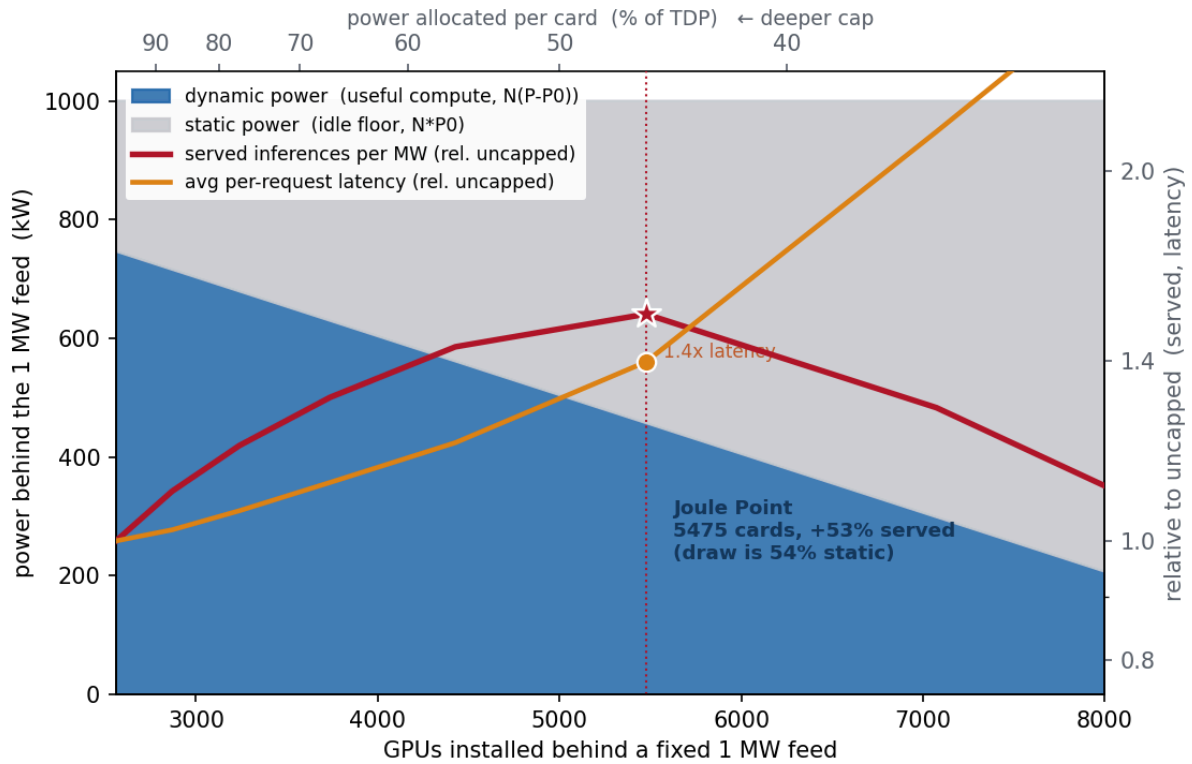


Figure A3. When power, not capital, binds, the JP is the interior optimum. Behind a fixed 1 MW feed (A100, ViT-B/32 at batch 64), installing more GPUs forces each to a deeper cap, shifting the megawatt from useful dynamic compute (blue) to the static idle floor (grey). Served inferences per megawatt (red) peak at the JP, near 5,500 installed cards, about 50 percent above the uncapped configuration. This is the capacity lever the Israel demonstration (WP4) operationalizes: under a rationed connection, efficiency becomes throughput. The same measured curves, allocated exactly across a fleet, dominate uncapped operation at every power budget from 100 down to 40 percent of total TDP; the preliminary measured-curve fleet simulation result appears in section 2.5.5.8.

Table A1. The five deliverables: what each is, who uses it

The project delivers five named artifacts, each with a defined user and use:

Deliverable	What it is	Who uses it, for what
ELF-2	An open dataset: power, throughput and latency at every operating point, board to wall, on current GPUs and production serving software	Researchers and vendors, as ground truth; every other deliverable is built and validated on it
The characterization kit	Software plus protocol: a one-test procedure and the two fitted models placing the JP of a new or unmeasured card within the validated hardware domain, in hours	A GPU engineer or operator qualifying a new card model or a new serving stack before deployment, at known cost
The control engine	An open-source control engine that assigns jobs and sets caps to keep a fleet inside its envelope, evaluated in the twin and demonstrated on a metered rented server before any deployment	An AI data-center manager, to cut energy per job and honor a committed envelope; its report doubles as the compliance record
The digital twin	A simulator replaying the measured curves of a described fleet and job mix, evaluating any policy or envelope scenario exactly before hardware is touched	An operator testing policies before deployment; a planner or regulator asking what a proposed facility could deliver and shed under a given connection
The grid interface and policy report	A blueprint: open formats for the envelope commitment, the flexibility offer and the delivery record, plus a regulator-facing report quantifying firm-plus-flexible connections against the national queue	The Electricity Authority and Noga, as the technical basis for offering conditional connections and auditing compliance

Table A1. The chain of use is direct: ELF-2 calibrates the models, the models feed the controller, the twin proves the controller before deployment, and the grid interface turns the controlled facility into something the regulator can contract with.